

Discussion. Beyond its primary purpose of clarifying the mechanisms through which GenAI adoption may differentially affect demand for junior and senior workers, the framework also helps reconcile two strands of evidence that appear to point in opposite directions: experimental studies consistently find that GenAI raises productivity most for less-experienced workers, while labor-market evidence, including ours, links GenAI adoption and exposure to declines in junior employment. The framework reconciles these findings by highlighting three margins that are largely outside the scope of experimental settings. First, experiments typically hold task assignment fixed, whereas junior and senior workers perform different tasks in practice, often with different exposure levels to GenAI. Second, experiments measure productivity gains on tasks that remain human-performed, but do not capture task displacement, whereby some tasks are fully automated and removed from human labor. Third, higher productivity need not translate into higher labor demand. In the fixed-output benchmark of the framework, when tasks are sufficiently complementary, productivity gains can reduce the number of junior workers required to perform a given set of tasks. More generally, the same force remains relevant whenever output expansion is not large enough to offset the labor-saving effect of higher productivity.

The remainder of the paper takes the framework to the data. Section 5.1 brings direct evidence on the displacement and reallocation channels—the productivity channel and the assumption $\delta_J \geq \delta_S$ are already well supported by the experimental literature. Then, Sections 5.2–5.5 focus on the framework’s central implication: how junior and senior employment within a firm change following GenAI adoption.

4 Data and Descriptive Patterns

4.1 Data Source and Sample

4.1.1 Primary Data Sources

Our primary data source is a detailed LinkedIn-based résumé dataset provided by Revelio Labs through WRDS. This dataset contains matched employer-employee information derived from individuals’ online profiles. For each worker, we observe all listed employ-

ment positions, including job titles, start and end dates, and the employing firm.¹⁶

A key feature of the dataset is the standardized *seniority level* variable for each position, constructed by Revelio through an ensemble modeling approach based on multiple sources of information. This measure combines information from (i) the worker’s current job (title, firm, and industry), (ii) their work history (tenure and previous seniority), and (iii) their age. These three inputs produce separate scores, which are averaged and then categorized into seven standardized seniority levels: Entry Level, Junior Level, Associate Level, Manager Level, Director Level, Executive Level, and Senior Executive Level. In the analysis that follows, we group positions into two broad categories: *juniors* (Entry and Junior) and *seniors* (Associate and above).¹⁷ Supplemental Appendices A.1 and A.2 provide comprehensive data validation; Section 4.3 below summarizes their main findings.

We merge the worker résumé data with Revelio’s job postings database at the firm level. The job postings database tracks firms’ recruitment activity since September 2021 and contains, for each posting, a firm identifier, posting date, and the raw text of the job description. We use these job-posting texts to construct our measure of firm-level GenAI adoption, as described in Section 4.2.

In addition, we incorporate the occupational GenAI exposure measures from Eloundou et al. (2024), merged with our position-level data via O*NET SOC codes. We rely on the GPT-4–based beta exposure measure (as in, for example, Brynjolfsson et al., 2025a), and classify all the positions into three categories: low exposure (0–25th percentile), medium exposure (25–75th percentile), and high exposure (75–100th percentile).

We also construct a representative sample of *tasks* from Revelio’s job postings database. The extraction pipeline is motivated by Hampole et al. (2025), who construct firm-level AI exposure by mapping résumé text to O*NET tasks. From the Revelio job-postings corpus we draw a representative sample of 1,000 firms containing 355,013 postings, and map each posting’s free-text description to a subset of the 19,265 standardized O*NET tasks using a two-stage pipeline. The first stage is a TF-IDF retriever that returns the top 50 candidate O*NET tasks for each posting; the second stage is a large language model

¹⁶The initial data download was performed in May 2025. In April 2026, we re-downloaded the positions data to capture profile updates that users added with a lag, ensuring more complete coverage of recent employment spells.

¹⁷More details on Revelio’s seniority classification methodology are available at <https://www.data-dictionary.reveliolabs.com/methodology.html#seniority>.

(OpenAI GPT-4o-mini) that reads the posting text together with the candidate list and selects the tasks it judges to be genuinely required. The procedure yields a panel of 2,748,186 posting-task pairs. Appendix A.14 describes the pipeline in detail, reports the exact prompt and JSON schema, and presents sanity checks on the output.

4.1.2 Main Sample

Our final sample merges all U.S. positions in the Revelio Labs dataset with job postings at the firm level. The resulting dataset covers 281,111 firms that were successfully matched to both employee position data and job postings, and that were actively hiring between January 2021 and March 2025.¹⁸ For these firms, we observe 161,192,969 positions held by 65,470,800 workers dating back to 2015, as well as 198,773,384 job postings since 2021, all with raw text descriptions.¹⁹

4.1.3 Measuring Workforce Dynamics by Seniority

We construct a monthly panel at the firm level. For each firm-by-month, we calculate the number of employees who held a position at the firm that started before and ended after that month, capturing the firm’s workforce size in that period. We repeat this calculation separately for each seniority category, enabling us to track the workforce composition over time. Additionally, we identify monthly inflows and outflows by seniority. For each firm-by-month, we define new hires as workers who began a new position at the firm that month, having most recently worked at another firm or for whom this is their first observed job. Separations are defined as workers whose position at the firm ended in that month and who either moved to a different firm or had no subsequent position listed. Finally, we define promotions as workers who start a new position at the firm after

¹⁸A firm is considered active if it had at least 20 new positions starting during this period.

¹⁹We exclude the top 1 percent of firms with the highest postings-to-hires ratios. Manual inspection indicates that many of these firms are HR intermediaries recruiting on behalf of other employers. In addition, we exclude the approximately 800 largest firms in the Revelio data (ranked by post-2021 new hires), representing about 0.2 percent of the firms in the sample, to avoid the influence of extreme outliers on our estimates. Supplemental Appendix A.19 shows that our main triple-difference results remain robust when these firms are added back.

previously holding a lower-seniority role within the same firm.^{20, 21}

4.2 GenAI Adoption

4.2.1 GenAI Integrator Vacancies

We identify GenAI adoption by detecting job postings that explicitly seek workers to implement or integrate GenAI technologies into firm workflows. Inspired by the approach of [Hampole et al. \(2025\)](#), we proceed in two steps. First, we compile a list of GenAI-related keywords and flag all postings containing at least one of them.²² Out of 198.8 million postings, 603,152 (0.30 percent) include at least one keyword. Second, we apply a large language model (LLM) classifier to this subset to identify genuine “GenAI integrator” postings—vacancies reflecting an active attempt to recruit workers tasked with adopting or implementing GenAI in the firm’s workflows—from generic GenAI mentions in job postings. (Supplemental Appendix [A.11.1](#) provides the exact prompt used). This process yields 131,845 postings, 0.066 percent of the full corpus, classified as GenAI integrator roles. Supplemental Appendix [A.12](#) provides a graphical overview of the procedure, and Supplemental Appendix [A.13](#) presents illustrative examples of postings classified as integrators and non-integrators.²³

²⁰Roughly 5 percent of positions are “contained positions,” meaning that another position for the same individual in the same firm fully overlaps their reported work period. We treat such positions as follows. If the container has lower seniority than the contained position, we shorten the container’s end date to the contained position’s start date, treating the latter as a promotion. If the container has higher seniority, we drop the contained position.

²¹Appendix [A.9](#) presents the aggregate time series of average junior and senior employment across firms. The two series expand in parallel through 2022 and then diverge: senior employment continues to grow while junior employment plateaus and begins to fall by mid-2023. This pattern is consistent with the findings of [Brynjolfsson et al. \(2025a\)](#), who document a comparable divergence in U.S. payroll data by worker age, and with the rising unemployment of recent college graduates documented in Supplemental Appendix [A.10](#).

²²Keywords include: Copilot, Claude, Gemini, large language model, LLM, generative AI, ChatGPT, Gen AI, GPT, LangChain, RAG, retrieval-augmented generation, vector embeddings, vector database, transformer-based model, prompt engineering, prompt design, LlamaIndex, Pinecone, Weaviate, Milvus, OpenAI API, Anthropic Claude API, Azure OpenAI, Google Vertex AI Generative, HuggingFace Transformers, and RetrievalQA.

²³As shown in Supplemental Appendix [A.11.1](#), we instructed the LLM to exclude AI producers. Upon manual inspection, however, we find that some vacancies are engaged in helping other firms integrate LLMs into their workflows, and we classify these postings as integrators. While this does not directly prove that such firms have embedded AI internally, we view it as a reasonable proxy, since firms offering integration services are highly likely to have adopted these technologies themselves.